

EASYSOURCING

Project Master Summary & Complete Build Phases

Version: 1.0 **Product:** Enterprise Asset Physical Verification, Audit & Reconciliation Platform

Business model: B2B SaaS + professional/managed asset audit services

Initial market: India

Primary goal: Digitize the complete physical asset verification lifecycle from asset-register intake to final client-approved audit report.

Important: This document is the consolidated project summary based on the EasySourcing concept discussed so far. It is intended to be used as a master reference before implementation and can be expanded after the second reference product is reviewed.

1. PROJECT VISION

- EasySourcing is not a basic inventory or CRUD asset-management application. It is a professional **Asset Physical Verification, Audit & Reconciliation Platform** combining field auditors, mobile technology, evidence capture, reconciliation, exception management, review, client approval and reporting.
- Core promise: **Every Asset. Verified. Reconciled. Accountable.**
- Business value: replace spreadsheets, paper checklists, manual photographs, disconnected GPS records, manual reconciliation and manual report preparation with one controlled digital workflow.

2. END-TO-END BUSINESS WORKFLOW

• Client → Asset Register → Import & Validation → Audit Project → Planning → Location Setup → Auditor Assignment → Mobile Field Verification → QR/Barcode/Search → Photo + GPS + Timestamp → Offline Sync → Reconciliation → Exception Detection → Exception Resolution → Quality Review → Client Review → Client Approval → Audit Finalization → Final Report → Archive.

3. PRODUCT SURFACES

- **Operations/Admin Web Portal:** used by EasySourcing internal teams for clients, assets, audits, locations, assignments, auditors, exceptions, evidence, reports, analytics and governance.
- **Auditor Mobile App:** optimized for fast field work: assignment, scan, verify, photo, GPS, exception capture, offline storage and synchronization.
- **Client Portal:** customer-facing dashboard for audit progress, assets, exceptions, evidence, reports and approvals.

4. TARGET USERS & ROLES

- **EasySourcing:** Super Admin, Operations Manager, Project Manager, Audit Manager, Reviewer, Auditor.
- **Client:** Client Admin, Client Manager, Client Viewer, Client Approver.
- Permissions must be granular and enforced server-side.

5. STRICT TECHNOLOGY STACK

- **Web:** Next.js 16+, React, TypeScript, App Router, Tailwind CSS, shadcn/ui, TanStack Query, React Hook Form, Zod.
- **Mobile:** React Native + Expo, TypeScript, SQLite/local persistent database, Expo Camera, Expo Location, SecureStore. Mobile is offline-first.
- **Backend:** NestJS + TypeScript, REST API, WebSocket only when genuinely required, OpenAPI/Swagger, server-side validation.
- **Database:** PostgreSQL + Prisma ORM.
- **Cache/Jobs:** Redis + BullMQ.
- **Files:** S3-compatible object storage for photos, evidence, reports, documents and import files.
- **Search:** PostgreSQL indexed/full-text search initially, with architecture that can later support OpenSearch.
- **Infrastructure:** Docker, Docker Compose for development, production containers, CI/CD, automated migrations, backups, monitoring and error tracking.
- **Architecture rule:** start as a modular monolith. Do not introduce microservices merely for fashion; keep clean module boundaries so high-load modules can later be extracted.

6. CORE BACKEND MODULES

- Auth; Tenants; Users; Roles; Permissions; Clients; Locations; Assets; Asset Tags; Audit Projects; Audit Assignments; Verification; Evidence; Reconciliation; Exceptions; Reviews; Approvals; Reports; Imports; Notifications; Analytics; Audit Logs; Settings.

7. MULTI-TENANCY & SECURITY

- Every tenant-owned record must be isolated using tenant context. Server-side authorization must prevent cross-tenant access.
- Security requirements: TLS, secure password hashing, JWT access tokens, refresh-token rotation, RBAC, API authorization, rate limiting, input validation, secure file handling, secure object URLs, audit logging, request IDs and secret management.
- Never trust a tenant ID or permission sent by the frontend.

8. DESIGN & UX DIRECTION

- Premium enterprise SaaS: professional, trustworthy, fast, data-dense but readable.
- Avoid toy-like cards, excessive gradients, excessive animation, random colors and legacy ERP appearance.
- Use strong typography, consistent spacing, subtle borders, clear status indicators, excellent tables, contextual actions, accessible components and responsive layouts.
- Admin/client web should be desktop-first but responsive. Auditor mobile should be a dedicated field experience rather than a shrunken admin portal.

9. PHASE 0 — FOUNDATION & ARCHITECTURE

- Create repository/monorepo, web app, mobile app, backend, database, shared types, validation, design system, environment configuration, Docker setup, CI foundation, logging, error handling and testing framework.

- Recommended monorepo: apps/web, apps/mobile, apps/api; packages/ui, packages/types, packages/config, packages/validation, packages/database.
- Delivery gate: project builds, types pass, lint passes, test framework works and basic deployment/health checks are validated.

10. PHASE 1 — AUTHENTICATION & ORGANIZATION

- Implement login, logout, password/session management, access and refresh tokens, user status, roles and permissions. Design for future MFA/SSO.
- Implement granular permissions such as client.read, asset.read, audit.create, audit.assign, audit.review, audit.finalize, exception.resolve, report.generate and user.manage.

11. PHASE 2 — CLIENT MANAGEMENT

- Client list, client profile, company information, contacts, industry, status, users, locations, assets, audits, reports and history.
- Support multiple clients under the EasySourcing organization.

12. PHASE 3 — LOCATION MANAGEMENT

- Configurable hierarchy: Client → Site → Building → Floor → Zone → Department → Room → Asset.
- Location fields: name, code, parent, address, GPS and status.
- Location dashboards show assets, verified, pending, exceptions and audit progress.

13. PHASE 4 — ASSET REGISTER & ASSET 360

- Asset identity: internal Asset ID, client Asset ID, description, category, subcategory, make, model, serial number, barcode and QR code.
- Financial: purchase date, purchase cost, capitalization date, current value and depreciation information.
- Physical: site, building, floor, zone, department, room, custodian, condition and status.
- Asset 360 tabs: Overview, Verification, History, Evidence, Documents, Exceptions, Transfers and Audit Trail.
- Lifecycle: Registered → Tagged → Assigned → Active → Maintenance → Missing/Disposed/Retired as appropriate.

14. PHASE 5 — IMPORT ENGINE

- Support XLSX and CSV.
- Workflow: Upload → Analyze → Map Columns → Validate → Preview → Confirm → Background Processing → Complete.
- Map client columns to EasySourcing fields. Detect duplicate IDs, duplicate serials, missing required fields, invalid dates/numbers, unknown locations and invalid statuses.
- Large imports must be asynchronous using object storage + BullMQ. Show progress and import results.

15. PHASE 6 — AUDIT PROJECT MANAGEMENT

- Create audit with client, project name, audit type, financial year, dates, asset register, locations, rules, evidence requirements, auditors and reviewers.
- Initial audit types: Annual Physical Verification, Fixed Asset Verification, Asset Tagging, Location Verification, Custodian Verification and Special Audit.
- Lifecycle: Draft → Planning → Ready → In Progress → Field Complete → Review → Client Review → Completed → Archived.

16. PHASE 7 — AUDIT PLANNING & ASSIGNMENTS

- Define scope by site, building, floor, zone, department, category, asset range or custom selection.
- Assign auditors by location or selected asset groups.
- Prevent accidental overlapping assignments and clearly surface conflicts.
- Operations view should show assigned, verified, pending, exceptions, completion and last sync.

17. PHASE 8 — AUDITOR MOBILE APP

- Mobile navigation: Home, Assignments, Scan, Exceptions, Profile.
- Home shows current assignment, verified, exceptions, pending, completion, offline queue and sync status.
- Primary action is SCAN ASSET.
- Fast verification target: a straightforward asset should normally be verified in roughly 10–20 seconds.

18. PHASE 9 — QR/BARCODE SCANNING

- Support QR, Code 128 and Code 39 initially.
- Flow: Scan → identify asset → load expected data → verify.
- If asset data is locally cached, scanning must work without network access.
- Provide manual search by Asset ID, serial, barcode, description, make and model when a tag cannot be scanned.

19. PHASE 10 — EVIDENCE CAPTURE

- Each verification can capture photographs, GPS, timestamp, auditor, device, result and remarks.
- Photos: camera capture, multiple images, compression, thumbnails, offline queue and upload retry.
- GPS: latitude, longitude, accuracy and timestamp. Clearly distinguish unavailable GPS from captured GPS; never fabricate coordinates.

20. PHASE 11 — OFFLINE-FIRST MOBILE

- Before field work, download only the auditor's assignment data. Do not download the entire client asset database.
- Offline capabilities: scan, search, verify, photograph, GPS, exceptions and remarks.
- Local verification must remain safe through app restarts and intermittent connectivity.

21. PHASE 12 — SYNC ENGINE

- Use local DB + sync queue + server validation + transactional persistence + confirmation.
- Every operation receives a unique operation ID. Sync must be idempotent, retryable, resumable and conflict-aware.
- Repeated sync requests must never create duplicate verification events.

22. PHASE 13 — SHEET-TO-FLOOR

- System provides expected assets. Auditor physically locates each asset and marks Verified, Missing, Exception or Deferred.
- Required evidence can be enforced by audit configuration.

23. PHASE 14 — FLOOR-TO-SHEET / DISCOVERY

- Auditor can discover physical assets not present in the register.
- Capture description, make, model, serial, location, condition, photo and remarks, then create an Unregistered Asset discovery/exception.

24. PHASE 15 — RECONCILIATION ENGINE

- Core comparison: registered/book data versus physical verification data.
- Authoritative result types: MATCHED, MISSING, LOCATION MISMATCH, CUSTODIAN MISMATCH, SERIAL MISMATCH, CONDITION EXCEPTION, UNREGISTERED, DUPLICATE and TAGGING ISSUE.
- Rules must be deterministic, transparent and unit-tested. AI must not decide authoritative financial/audit results.

25. PHASE 16 — EXCEPTION CENTER

- Centralized exception management with filters for client, audit, location, department, category, severity, auditor, status and date.
- Types: Missing, Location Mismatch, Custodian Mismatch, Serial Mismatch, Damaged, Unregistered, Duplicate and Tag Issue.
- Lifecycle: Open → Assigned → Investigating → Resolved → Reviewer Review → Approved → Closed.

26. PHASE 17 — EXCEPTION RESOLUTION

- Capture action taken, explanation, updated value, evidence, resolver and timestamp.
- Do not overwrite the original audit event. Corrections should create controlled new events.

27. PHASE 18 — QUALITY REVIEW & RE-VERIFICATION

- Reviewer can inspect verification, photographs, GPS and remarks; reject findings, request re-verification, approve exceptions and approve audit.

- Re-verification creates a new controlled event while preserving the original history.

28. PHASE 19 — OPERATIONS DASHBOARD

- Show active audits, assets in scope, verified, exceptions, overall progress, auditor status, assignment progress, location progress and synchronization state.
- Near-real-time updates may use WebSockets where genuinely useful.

29. PHASE 20 — CLIENT PORTAL

- Client navigation: Dashboard, Audits, Assets, Exceptions, Evidence, Reports, Approvals.
- Client sees only its organization. Internal EasySourcing notes remain private unless explicitly exposed.
- Client can review evidence and, according to permissions, comment, request clarification, accept/reject or approve.

30. PHASE 21 — EVIDENCE CENTER

- Centralized evidence library searchable by audit, asset, exception, location, auditor and date.
- Every evidence item remains linked to its originating verification/event.

31. PHASE 22 — REPORTING ENGINE

- Reports: Executive Summary, Complete Asset Verification, Missing Assets, Location Mismatch, Custodian Mismatch, Serial Mismatch, Damaged Assets, Unregistered Assets, Duplicate Assets, Evidence Report, Location-wise, Department-wise and Auditor Productivity.
- Reports should be professional consulting/audit deliverables, not raw database dumps.
- Large report generation and exports run asynchronously.

32. PHASE 23 — REPORT VERSIONING & FINALIZATION

- Support Draft, Version 1, Version 2 and Final.
- Record generator, timestamp, data snapshot and approval state.
- After finalization, freeze audit results/evidence associations and require controlled amendment for later corrections.

33. PHASE 24 — DIGITAL AUDIT TRAIL

- Record WHO, WHAT, WHEN, WHERE, BEFORE and AFTER for important actions.
- Audit records should be append-oriented and protected from ordinary modification.

34. PHASE 25 — ANALYTICS & RISK

- Metrics: verification rate, exception rate, missing rate, location accuracy, custodian accuracy, condition distribution, auditor productivity and audit completion.

- Risk indicators may be configurable and transparent. Example: location-level high/medium/low risk based on documented rules.

35. PHASE 26 — QR/TAG MANAGEMENT

- Generate QR/barcodes, bulk tags and printable PDFs. Track tag replacement/history.
- Initial priority: QR + barcode + mobile camera. Architecture remains ready for NFC/RFID later.

36. PHASE 27 — NOTIFICATIONS

- In-app and email notifications for assignment, deadlines, audit delays, exception assignment/resolution, re-verification, review, client approval and report readiness.
- Use background jobs.

37. PHASE 28 — SEARCH, BULK ACTIONS & EXPORT

- Global search across Asset ID, serial, barcode, QR, description, make, model, location and custodian.
- Bulk operations: assign, reassign, export, resolve and status changes, with confirmations for dangerous actions.
- Exports: XLSX, CSV and PDF. Large exports are background jobs.

38. PHASE 29 — INSTRUMENT MANAGEMENT FOUNDATION

- Keep a separate bounded domain for instruments/calibration.
- Fields can include instrument ID, type, manufacturer, model, serial, measurement range, accuracy, calibration status, last/next calibration, certificate and provider.
- Future workflow: Instrument → Calibration → Certificate → Result → Approval → Next Due Date.

39. PHASE 30 — ERP & EXTERNAL INTEGRATIONS

- Create adapters for future SAP, Oracle, Tally, HR systems, SSO and external APIs.
- Do not hard-code an ERP into the core asset/audit domain.

40. PHASE 31 — PERFORMANCE & SCALABILITY

- Target architecture should support 1M+ assets per tenant without loading the full dataset into browser memory.
- Use indexed queries, server-side filtering, pagination/cursor pagination, virtualization, caching, background jobs and streaming exports.
- Target normal API p95 under 500ms and simple locally cached asset lookup under about 1 second, subject to production load testing.
- Never process large files synchronously or re-download/re-upload unchanged mobile data unnecessarily.

41. PHASE 32 — OBSERVABILITY

- Structured application logs, metrics, health/readiness checks, queue monitoring and error tracking.
- Track API latency, errors, DB performance, queue latency, sync failures, import failures, report failures and storage failures.
- Use Sentry or an equivalent production error-monitoring system.

42. PHASE 33 — BACKUP & DISASTER RECOVERY

- Automated backups, point-in-time recovery and regular backup restoration tests.
- Target RPO ≤ 15 minutes and RTO ≤ 1 hour, subject to final infrastructure configuration.

43. PHASE 34 — TESTING

- Unit tests for reconciliation, permissions, imports, state transitions, sync and conflicts.
- Integration tests for authentication, DB, API, imports, evidence and reports.
- E2E test the complete Client $\rightarrow$ Register $\rightarrow$ Audit $\rightarrow$ Assignment $\rightarrow$ Mobile Verification $\rightarrow$ Evidence $\rightarrow$ Reconciliation $\rightarrow$ Exception $\rightarrow$ Resolution $\rightarrow$ Review $\rightarrow$ Approval $\rightarrow$ Final Report workflow.
- Mobile tests must include online, offline/airplane mode, poor network, restart, retry, duplicate sync, camera, GPS, QR, barcode and large assignments.

44. PHASE 35 — PRODUCTION HARDENING

- Perform security testing, load testing, database performance testing, large Excel testing, offline testing, poor-network testing, large-image testing, report testing, disaster-recovery testing and cross-browser/device testing.
- Production release requires all critical quality gates to pass.

45. DATABASE CORE ENTITIES

- Tenant, User, Role, Permission, Client, ClientUser, Location, LocationType, Asset, AssetCategory, AssetTag, AuditProject, AuditScope, AuditAssignment, Verification, VerificationEvent, Evidence, Photo, Document, Exception, ExceptionComment, ExceptionResolution, Review, Approval, Report, ReportVersion, ImportJob, ImportRow, Notification and AuditLog.

46. API ARCHITECTURE

- Version APIs under /api/v1/. Use authentication, authorization, validation, consistent errors and request IDs.
- Examples: POST/GET clients; POST assets/import; GET assets/:id; POST/GET audits; POST assignments; POST verifications; POST evidence; GET/PATCH exceptions; POST reports; POST approvals.
- Never expose stack traces or internal implementation details to clients.

47. DATA INTEGRITY RULES

- Use database transactions for critical operations, proper foreign keys, constraints and indexes.
- Use soft deletion where appropriate. Do not casually delete finalized audit evidence.

- Once an audit is finalized, original verification events remain historically traceable and are not overwritten.

48. MASTER UI STRUCTURE

- **Admin:** Overview; Clients; Locations; Audits; Assets; Operations; Exceptions; Evidence; Reports; Analytics; Team; Settings; Audit Logs.
- **Mobile:** Home; Assignments; Scan; Exceptions; Profile.
- **Client:** Dashboard; Audits; Assets; Exceptions; Evidence; Reports; Approvals.

49. DELIVERY METHOD FOR GLM/Z.AI AGENT

- Do not ask the coding agent to build the entire system blindly in one pass.
- Execution order: Understand → Plan → Implement → Test → Review → Optimize → Document → Commit.
- Complete and verify each phase before starting the next. Preserve working functionality and maintain a regression suite.
- Before changing an existing workflow, inspect its current behavior, add/update tests, implement the change, run regression tests and verify affected web/mobile flows.

50. NON-NEGOTIABLE DEVELOPMENT RULES

- Build production software, not a demo. Do not change the locked stack without approval. Do not remove existing functionality. Do not use mock data for production workflows. Do not allow cross-tenant access. Do not process large files synchronously. Do not load massive datasets into browser memory. Do not store large media in PostgreSQL. Do not require internet for field verification. Do not silently overwrite conflicts. Do not delete finalized audit evidence. Do not make AI authoritative for audit results. Every critical business rule must be tested. Every schema change requires a migration. Every protected API requires server-side authorization. Every mobile verification must work offline. Every audit event must be traceable.

51. PHASE COMPLETION GATE

- Every phase must finish with: build success, TypeScript/type check success, lint success, unit tests, integration tests where applicable, E2E/critical workflow verification where applicable, database migration verification, security/authorization checks and regression testing.
- Do not declare a phase complete merely because the UI appears on screen.

52. MVP / FIRST COMMERCIAL RELEASE

- The first commercially useful release should include Foundation, Auth, Clients, Locations, Asset Register, Import Engine, Audit Projects, Assignments, Auditor Mobile, QR/barcode, verification, photo/GPS, offline operation, sync, Sheet-to-Floor, Floor-to-Sheet discovery, reconciliation, exceptions, review and essential professional reporting.
- This creates the full differentiating loop: **Client** → **Register** → **Audit** → **Assign** → **Mobile Verify** → **Evidence** → **Reconcile** → **Exceptions** → **Review** → **Report**.

53. LONG-TERM PRODUCT DIRECTION

- After the core platform is stable, expand into advanced analytics, instruments/calibration, ERP integrations, RFID/NFC, AI-assisted OCR/photo classification/anomaly suggestions and additional audit templates.
- AI should remain assistive. Human-controlled and deterministic audit workflows remain authoritative.

54. FINAL PRODUCT STATEMENT

- **EasySourcing transforms physical asset verification from manual spreadsheets into a controlled digital workflow covering planning, field verification, evidence capture, offline operation, reconciliation, exception management, review, client approval and professional final audit reporting.**